THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME — SET A

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 %	$\frac{10}{100} \times 100 = 10$
2	**(a)** 5 %	$\frac{5}{100} \times 100 = 5$
3	**(c)** solution	uniform composition
4	**(c)** a saturated solution	definition
5	**(b)** fractional distillation	small differences in boiling points
6	**(d)** naphthalene	solid → vapour directly
7	**(a)** Tyndall effect	light path becomes visible
8	**(d)** an alloy	e.g. brass
9	**(c)** 0.9 % m/v	sodium chloride in water
10	**(b)** distillation	capturing the fragrance of rain on soil

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The amount of solute dissolved in a given amount of solvent or solution (1). Mass by mass %, mass by volume %, volume by volume % (1).	1+1
12	Mass of solution $= 100$ g; $\frac{10}{100} \times 100 = 10\%$ (1+1).	1+1
13	$\frac{5}{100} \times 100 = 5\%$ m/v.	1+1
14	Solubility of solids generally increases with temperature and that of gases decreases, so the value depends on temperature (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mass $= \frac{4}{100} \times 300 = 12$ g (3).	1+1+1
16	Volume of concentrate $= 30$ mL (1); $\frac{30}{150} \times 100$ (1) $= 20\%$ v/v (1).	1+1+1
17	$\frac{5}{100} \times 200 = 10$ mL of glacial acetic acid (2), made up to 200 mL with water (1).	2+1
	OR: $\frac{0.9}{100} \times 500 = 4.5$ g (3).	1+1+1
18	Solubility (g per 100 g water) against temperature (1). Inferences: B dissolves much more in hot water than cold (1); cooling a hot saturated solution of B deposits a lot of crystals (1).	1+1+1
	OR: Filtering removes insoluble impurities (1.5); the acid prevents unwanted reactions and gives pure crystals (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) g of solute per 100 g solution; g per 100 mL solution; mL per 100 mL solution (2). (ii) $\frac{25}{250} \times 100 = 10\%$ (1.5). (iii) $\frac{4}{100} \times 250 = 10$ g (1.5).	2+1½+1½
	OR: (i) Too little does not protect the crop and too much can harm it (1). (ii) $\frac{1}{100} \times 100 = 1\%$ v/v (1). (iii) 20 L = 20000 mL; 1% = 200 mL (2). (iv) A homogeneous mixture (solution) (1).	1+1+2+1
20	(i) Formation of pure crystals from a hot saturated solution on cooling (1). (ii) Saturate hot water with copper sulfate plus a few drops of dilute sulfuric acid, filter hot, cool slowly, filter the crystals, rinse and dry; precautions — copper sulfate is toxic, wear gloves, teacher adds acid (3). (iii) Snowflakes, frost, candy sugar; crystals in mineral deposits (1).	1+3+1
	OR: (i) B (the curve is higher) (1). (ii) B, because its solubility drops sharply on cooling (2). (iii) B, because large differences in solubility with temperature give a better yield (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 20 %	$\frac{20}{100} \times 100 = 20$
2	**(d)** 0.9 %	$\frac{9}{1000} \times 100 = 0.9$
3	**(b)** water	the substance present in larger amount that dissolves the solute
4	**(c)** crystallization	differences in solubility with temperature
5	**(d)** paper chromatography	different components move at different speeds
6	**(a)** solid carbon dioxide that sublimes	used for storing ice cream
7	**(d)** milk diluted with water	colloids scatter light
8	**(d)** melting them together and cooling	solid solution on cooling
9	**(c)** colour and writing	chroma + graphein
10	**(c)** gives pure crystals and removes soluble impurities	uses cooling of a saturated solution

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A solution that cannot dissolve any more solute at that temperature (2).	2
12	Formation of crystals of a pure solid from a hot saturated solution on cooling (1). Principle: difference in solubility of a substance at different temperatures (1).	1+1
13	Slow cooling gives particles enough time to arrange to form large, pure, well-formed crystals (2).	2
14	By distillation (1); acetone boils at about 56 °C, water at 100 °C, so acetone vaporises first and is condensed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make a saturated solution in hot water with a few drops of dilute acid (1); filter the hot solution (1); cool slowly (½); filter crystals, rinse and dry (½).	1+1+1
16	Mixture heated in a flask; lower-boiling liquid vaporises (1); vapour passes through a condenser and condenses (1); distillate collected separately; used to recover solvent/extract fragrance (1).	1+1+1
17	Distillation: acetone vaporises first and is condensed (2). Evaporation would let both vaporise and lose the liquid (1).	2+1
	OR: A method for separating components whose boiling points differ by less than about 25 °C using a fractionating column (1.5); used for crude petroleum into petroleum gas, petrol, kerosene, diesel (1.5).	1½+1½
18	Spot on a line drawn 2 cm from the bottom (1); strip dipped in water with the spot above the level (1); water rises and carries the dyes at different speeds, separating them (1).	1+1+1
	OR: Components differ in how strongly they interact with the paper and the solvent (2); e.g. the dyes of black ink spread as different coloured bands (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Simple: liquids with boiling points differing by more than ~25 °C; fractional: small differences, uses a fractionating column (2). (ii) Flask, thermometer, condenser with water inlet/outlet, receiver (2). (iii) Petroleum gas (LPG as cooking fuel), petrol (vehicle fuel), kerosene, diesel (1).	2+2+1
	OR: (i) Traditional distillation (1). (ii) The volatile fragrance components are vapourised by heating, then cooled in a condenser and collected (2). (iii) Recovering a solvent, separating liquids, fragrance extraction from flowers (1). (iv) The fragrance is dissolved/evaporating, not a solid that can be retained on a filter (1).	1+2+1+1
20	(i) Separation of mixture components based on different speeds on a paper (1). (ii) 3 cm strip, pencil line 2 cm from the bottom, ink spot at centre, dip in water with the spot above the water level (2). (iii) Water rises carrying the dyes; different colours separate at different heights as they interact differently with paper and solvent (1). (iv) Otherwise the spot would dissolve into the solvent instead of travelling (1).	1+2+1+1

	OR: (i) Solution: salt in water (clear, no settling); colloid: milk, fog (scatter light, stable); suspension: muddy water (settles) (3). (ii) Scattering of light by colloidal particles (1). (iii) Pass a torch beam through a dilute milk sample; the beam path is visible (1).	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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